

## Questions

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## Q1 - 2024 (04 Apr Shift 1)

If a rubber ball falls from a height  $h$  and rebounds upto the height of  $h/2$ . The percentage loss of total energy of the initial system as well as velocity ball before it strikes the ground, respectively, are :

(1) 50%,  $\sqrt{2gh}$

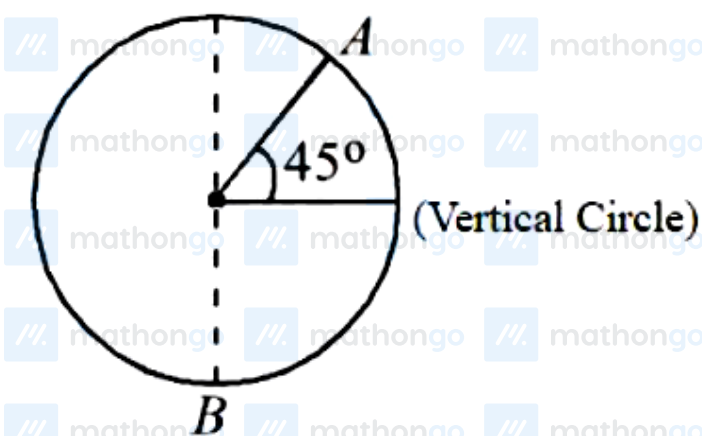
(2) 50%,  $\sqrt{gh}$

(3) 40%,  $\sqrt{2gh}$

(4) 50%,  $\sqrt{\frac{gh}{2}}$

## Q2 - 2024 (04 Apr Shift 2)

A body of  $m$  kg slides from rest along the curve of vertical circle from point  $A$  to  $B$  in friction less path. The velocity of the body at  $B$  is:



(given,  $R = 14$  m,  $g = 10$  m/s<sup>2</sup> and  $\sqrt{2} = 1.4$  )

(1) 16.7 m/s

(2) 19.8 m/s

(3) 10.6 m/s

(4) 21.9 m/s

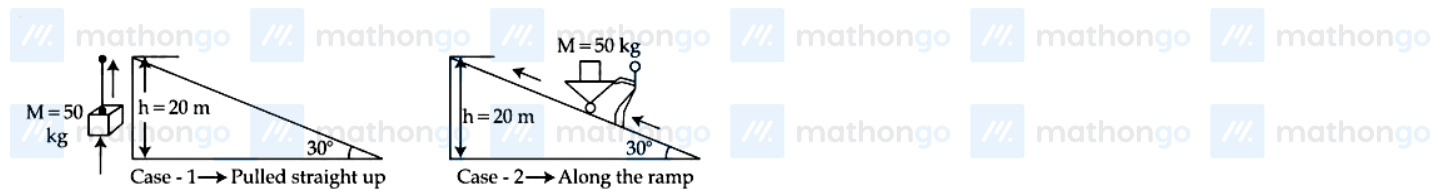
## Q3 - 2024 (05 Apr Shift 1)

A body of mass 50 kg is lifted to a height of 20 m from the ground in the two different ways as shown in the figures. The ratio of work done against the gravity in both the respective cases, will be :

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(1)  $1 : 2$

(2)  $\sqrt{3} : 2$

(3)  $2 : 1$

(4)  $1 : 1$

## Q4 - 2024 (05 Apr Shift 2)

A body is moving unidirectionally under the influence of a constant power source. Its displacement in time  $t$  is proportional to :

(1)  $t$

(2)  $t^{3/2}$

(3)  $t^2$

(4)  $t^{2/3}$

## Q5 - 2024 (06 Apr Shift 1)

A bullet of mass 50 g is fired with a speed 100 m/s on a plywood and emerges with 40 m/s. The percentage loss of kinetic energy is :

(1) 84%

(2) 16%

(3) 32%

(4) 44%

## Q6 - 2024 (08 Apr Shift 1)

A stationary particle breaks into two parts of masses  $m_A$  and  $m_B$  which move with velocities  $v_A$  and  $v_B$  respectively. The ratio of their kinetic energies ( $K_B : K_A$ ) is :

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(1)  $v_B : v_A$

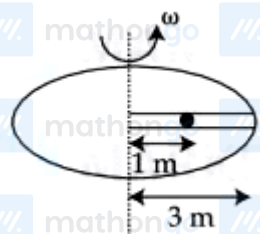
(2)  $m_B : m_A$

(3)  $m_B v_B : m_A v_A$

(4)  $1 : 1$

## Q7 - 2024 (08 Apr Shift 2)

A circular table is rotating with an angular velocity of  $\omega$  rad/s about its axis (see figure). There is a smooth groove along a radial direction on the table. A steel ball is gently placed at a distance of 1 m on the groove. All the surfaces are smooth. If the radius of the table is 3 m, the radial velocity of the ball w.r.t. the table at the time ball leaves the table is  $x\sqrt{2}\omega$  m/s, where the value of  $x$  is \_\_\_\_\_.



## Q8 - 2024 (09 Apr Shift 1)

A particle of mass  $m$  moves on a straight line with its velocity increasing with distance according to the equation  $v = \alpha\sqrt{x}$ , where  $\alpha$  is a constant. The total work done by all the forces applied on the particle during its displacement from  $x = 0$  to  $x = d$ , will be :

(1)  $\frac{m}{2\alpha^2 d}$

(2)  $\frac{md}{2\alpha^2}$

(3)  $2 m\alpha^2 d$

(4)  $\frac{m\alpha^2 d}{2}$

## Q9 - 2024 (09 Apr Shift 2)

A force  $(3x^2 + 2x - 5)$  N displaces a body from  $x = 2$  m to  $x = 4$  m. Work done by this force is \_\_\_\_\_ J

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## Answer Key

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**Q1** (1) mathongo // math // **Q2** (4) // mathongo **Q3** (4) mathongo // math // **Q4** (2) // mathongo

**Q5** (1) mathongo // math // **Q6** (1) // mathongo **Q7** (2) mathongo // math // **Q8** (4) // mathongo

**Q9** (58) thongo // mathongo // mathongo // mathongo // mathongo // mathongo

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## Solutions

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Q1

$$\text{Velocity just before collision} = \sqrt{2gh}$$

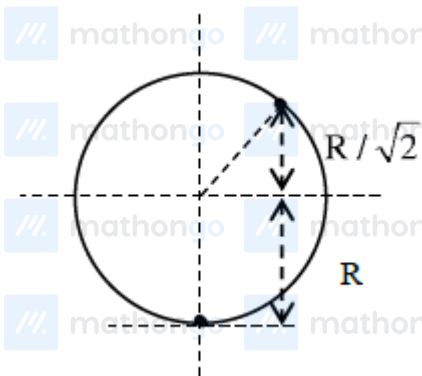
$$\text{Velocity just after collision} = \sqrt{2g\left(\frac{h}{2}\right)}$$

$$\begin{aligned}\therefore \Delta KE &= \frac{1}{2}m(2gh) - \frac{1}{2}mgh \\ &= \frac{1}{2}mgh\end{aligned}$$

 $\therefore$  % loss in energy

$$= \frac{\Delta KE}{KE_i} \times 100 = \frac{\frac{1}{2}mgh}{\frac{1}{2}mg2h} \times 100 = 50\%$$

Q2



Apply W.E.T. from A to B

$$\Rightarrow W_{mg} = K_B - K_A$$

$$\Rightarrow mg \times \left( \frac{R}{\sqrt{2}} + R \right) = \frac{1}{2}mv_B^2 - 0 \{v_A = 0 \text{ rest}\}$$

$$\Rightarrow mgR \frac{(\sqrt{2} + 1)}{\sqrt{2}} = \frac{1}{2}mv_B^2$$

$$\Rightarrow \sqrt{gR \frac{2(\sqrt{2} + 1)}{\sqrt{2}}} = v_B$$

$$\Rightarrow \sqrt{\frac{10 \times 14 \times 2(2.4)}{1.4}} = v_B$$

$$\Rightarrow 21.9 = v_B$$

Q3

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## Solutions

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Work done by gravity is independent of path. It depends only on vertical displacement so work done in both cases will be same.

Q4

$$P = \text{constant} \Rightarrow FV = \text{constant}$$

$$\Rightarrow m \frac{dV}{dt} V = \text{constant}$$

$$\int_0^V V dV = (C) \int_0^t dt$$

$$\left( \frac{V^2}{2} \right) = Ct$$

$$V \propto t^{1/2}$$

$$\frac{ds}{dt} \propto t^{1/2}$$

$$\int_0^S ds = K \int_0^t t^{1/2} dt$$

$$S = K \times \frac{2}{3} t^{3/2}$$

$$S \propto t^{3/2}$$

$\therefore$  displacement is proportional to  $(t)^{3/2}$

Q5

$$K_i = \frac{1}{2} m(100)^2$$

$$K_f = \frac{1}{2} m(40)^2$$

$$\% \text{ loss} = \frac{|K_f - K_i|}{K_i} \times 100$$

$$= \frac{\left| \frac{1}{2} m(40)^2 - \frac{1}{2} m(100)^2 \right|}{\frac{1}{2} m(100)^2} \times 100$$

$$= \frac{|1600 - 100 \times 100|}{100} = 84\%$$

Q6

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## Solutions

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Initial momentum is zero.

$$\text{Hence } |P_A| = |P_B|$$

$$\Rightarrow m_A v_B = m_B v_B$$

$$\frac{(KE)_A}{(KE)_B} = \frac{\frac{1}{2} m_A v_A^2}{\frac{1}{2} m_B v_B^2} = \frac{v_A}{v_B}$$

$$\frac{(KE)_B}{(KE)_A} = \frac{v_B}{v_A}$$

Q7

$$a_c = \omega^2 x$$

$$\frac{v dv}{dx} = \omega^2 x$$

$$\int_0^v v dv = \int_1^3 \omega^2 x dx$$

$$\frac{v^2}{2} = \omega^2 \left[ \frac{x^2}{2} \right]$$

$$\frac{v^2}{2} = \frac{\omega^2}{2} [3^2 - 1^2]$$

$$v = 2\sqrt{2}\omega$$

$$x = 2$$

Q8

$$v = \alpha\sqrt{x}$$

$$\text{at } x = 0 : v = 0$$

$$\& \text{ at } x = d; v = \alpha\sqrt{d}$$

$$W.D = K_f - K_i$$

$$W.D = \frac{1}{2} m(\alpha\sqrt{d})^2 - \frac{1}{2} m(0)^2$$

$$\Rightarrow W.D = \frac{m\alpha^2 d}{2}$$

Q9

$$W = \int_{x_1}^{x_2} F dx$$

$$W = \int_2^4 (3x^2 + 2x - 5) dx$$

$$W = [x^3 + x^2 - 5x]_2^4$$

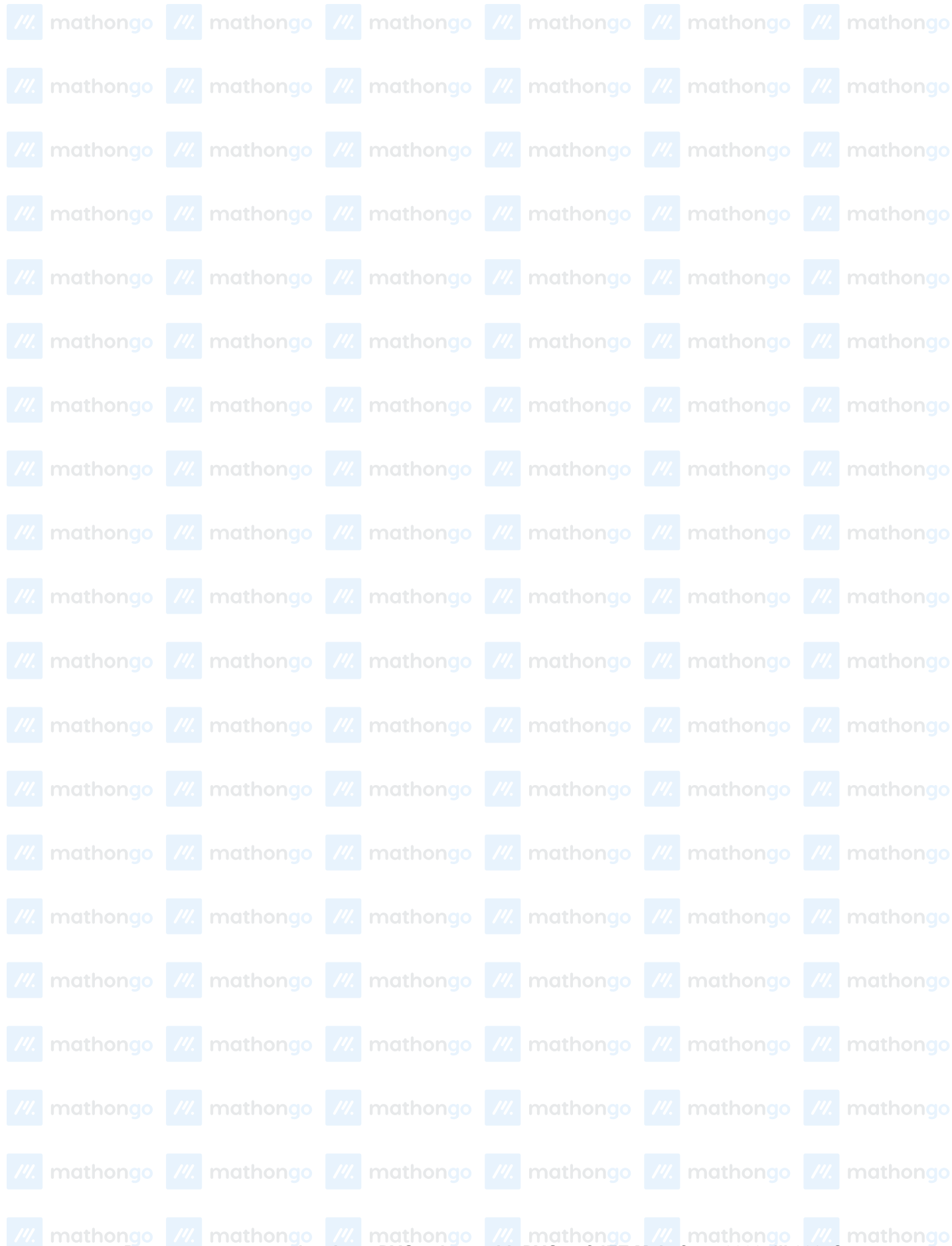
$$W = [60 - 2]J = 58J$$

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Solutions

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